ALISA OMELCHENKO

347-570-5722 | alisa.a.omelchenko@gmail.com | linkedin.com/in/alisa-omelchenko | GitHub @AlisaOmel
alisaomel.github.io | Pittsburgh, PA, 15207

RESEARCH PROFILE

PhD candidate in the CMU-Pitt joint Computational Biology program focused on creative problem-solving. My goal is to work collaboratively to develop accessible tools that empower biologists and move science forward. With hands-on experience across both wet-lab and computational research, I bridge the gap between experimental biology and computer science. My work spans multi-omic network analysis, protein modeling, and agentic AI applications in immunology and disease. I have first-author publications designing machine learning methods and network approaches to analyze single-cell transcriptomic and Multiome data. Beyond the bench and the laptop, I communicate biological concepts to audiences ranging from scientists to policymakers, through mentoring, science policy advocacy, and cross-disciplinary collaboration. I'm drawn to fast-paced environments where I can partner directly with research teams to turn open biological questions into discoveries with clinical relevance.

EDUCATION

CMU-Pitt Joint Computational Biology: Pittsburgh, Pennsylvania *Sept 2022- Present*
PhD Candidate in Computational Biology *Expected Graduation 2026-2027*

NYU Tandon School of Engineering: Brooklyn, New York *Graduated May 2018*
M.S. Biotechnology *GPA: 3.94*
B.S. BioMolecular Sciences (Magna Cum Laude) *GPA: 3.61*

SKILLS

Computer Languages: Python, R, Bash
Computational Skills: HPC (Slurm, LSF), protein and interaction language models (LLMs, pLMs/iLMs), interpretable ML, Agentic AI for Biological Applications (Biomni, Co-Scientist), multi-omic and sc-omic network analysis, protein structure modeling and visualization (AlphaFold3/Boltz2, OpenMM, PyMOL, fpocket), processing imaging data (ImageJ/FIJI, fMRI)
Laboratory Skills: Genotyping/PCR, ELISA, flow cytometry, calcium imaging, confocal microscopy, molecular cloning, animal handling (mice, *Drosophila*)

RESEARCH EXPERIENCE

St. Jude Center for Applied Bioinformatics: Summer Research Intern *June 2026-Present*
Advised by: Gang Wu and Jeremy Chase Crawford
- Prioritized patient-specific cancer neo-epitopes and variants from NGS data (WTS, WGS, WES) for a B-ALL cohort, identifying patterns associated with relapse and subtypes for future clinical translation.
- Predicted WT and mutant protein complex structures (AlphaFold3/Boltz2) and ran MD simulations (OpenMM) to assess interaction stability and characterize variant effects on protein function.
- Directed agentic AI (Biomni, Co-Scientist) and LLMs via prompt engineering to explore testable hypotheses and design validation strategies, then evaluated outputs for biological accuracy.
- Supervised 2 high school interns in the Research Immersion Program, guiding them through hands-on research.

CMU-Pitt Joint Computational Biology: Graduate Student *Sept 2022-Present*
Advised by: Jishnu Das
- Developed SWING, a first-in-class interaction-based vocabulary language model (iLM) to predict peptide:peptide interactions and variant perturbations collaboratively (Published in Nat. Methods).
- Integrated multi-omic datasets and used a systems biology network-based approach to identify a core T follicular helper (Tfh) cell network that is consistent across contexts (Publisehd in Nat. Commun.).
- Collaborated to identify novel mechanisms of HIV infection in Tfh cells. (In Revision at Nat. Commun.).
- Extended iLMs to capture ternary (pMHC:TCR) and non-peptide partners (drug:protein) interactions, enabling generalizable, patient-specific predictions for clinical translation.

Virginia Tech School of Neuroscience Ni Lab: Laboratory Manager *July 2019-July 2022*
Advised by: Lina Ni
- Created an ImageJ plugin to automate data extraction and processing of experiments (Published in JoVE).
- Identified the function, molecular components, and properties in the thermosensory pathway of *Drosophila* larvae (Published in JEB & Front. Mol. Neurosci.).
- Designed automated workflows for fly tracking to ensure reproducible results and speed up data analysis (JEB).

• Generated new fly strains for analysis of gene function through molecular cloning and recombination.

Landos Biopharma: Laboratory Technician July *2018-July 2019*
Advised by: *Josep Bassaganya-Riera*
• Identified the mechanism of action for drugs through evaluation of differentially expressed gene (DEG) patterns.
• Collected and analyzed data from necropsies, cell cultures, flow cytometry, ELISA, and western blots.
• Handled, monitored, and performed routine medical treatments on mice, rats, and pigs.

NYU Langone Health: Volunteer Researcher *April 2018-June 2018*
Advised by: Timothy J. Cardozo
• Designed structure and evaluated stability of antibody eliciting epitopes for staph database and staph vaccines.
• Analyzed alignments to predict protein structures through homology modeling of proteins not in the PDB.

Nathan Kline Institute: Research Assistant *September 2016-May 2018*
Advised by: Cameron R. Craddock
• Determined functional similarities of the brain between healthy individuals with comparable personality traits.
• Mapped cluster differences of fMRI data on brain templates to visualize subgroup differences in activation.
• Assessed the quality of 225 skull-stripped brain images and manually edited them.

Child Mind Institute: Volunteer Research Assistant *September 2016-May 2018*
Advised by: Steven Giavasis
• Improved the speed, usability, and statistical implementation of Python scripts used by the institute.
• Enhanced a Python script to utilize a Docker container and parallelize the data processing increasing the speed by 30%. The script is included in the current version of C-PAC software.

PUBLICATIONS

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. Nat Commun 17, 9099 (2026).

Del Rio Estrada, P. M.*, Georgakis, S.*, Orfanakis, M., **Omelchenko, A.**, Coirada, F., Dos Santos, J., ... & Ribeiro S.P (2026). IL-10-and TGF-β-Driven BCL6 Expression Suppresses Antiviral Defenses and Renders Lymph Node T Follicular Helper Cells Permissive to HIV Infection. *Preprint*. [In Revision at Nature Communications]

Omelchenko, A. A.*, Siwek, J. C.*, Chhibbar, P.*, Arshad, S., Rosengart, A., Nazarali, I., ... & Das, J. (2025). Sliding Window Interaction Grammar (SWING): a generalized interaction language model for peptide and protein interactions. *Nature Methods*, 1-13.

Omelchenko, A. A., Bai, H., Hussain, S., Tyrrell, J. J., Klein, M., & Ni, L. (2022). TACI: An ImageJ Plugin for 3D Calcium Imaging Analysis. JoVE (Journal of Visualized Experiments), (190), e64953.

Omelchenko, A. A., Bai, H., Spina, E. C., Tyrrell, J. J., Wilbourne, J. T., & Ni, L. (2022). Cool and warm ionotropic receptors control multiple thermotaxes in Drosophila larvae. *Frontiers in Molecular Neuroscience*.

Huda, A.*, **Omelchenko, A. A.***, Vaden, T. J.*, Castaneda, A. N., & Ni, L. (2022). Responses of different Drosophila species to temperature changes. *Journal of Experimental Biology*, *225*(11), jeb243708.

Tyrrell, J. J., Wilbourne, J. T., **Omelchenko, A. A.**, Yoon, J., & Ni, L. (2021). Ionotropic Receptor-dependent cool cells control the transition of temperature preference in Drosophila larvae. *PLoS genetics*, *17*(4), e1009499. https://doi.org/10.1371/journal.pgen.1009499

Wang, X., Li, X. H., Cho, J. W., Russ, B. E., Rajamani, N., **Omelchenko, A.**, Ai, L., Korchmaros, A., Sawiak, S., Benn, A.R., Garcia-Saldivar, P., Wang, Z., Kalin N.H., Schroeder, C.E., Craddock, R.C., Fox, A.S., Evans, A.C., Messinger, A., Milham, M.P, Xu, T. (2021). U-Net Model for Brain Extraction: Trained on Humans for Transfer to Non-human Primates. *NeuroImage*, 118001. https://doi.org/10.1016/j.neuroimage.2021.118001

*These authors contributed equally

CONFERENCES and AWARDS

Presentations:

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: 25th Annual Immunology Retreat, October 2026.

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: AAI, April 2026.

Computational approaches to understand immune system modulation through protein interactions. Sip of Science, December 2025.

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: Center for Systems Immunology Retreat, November 2025

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: ISMB/ECCB, July 2025.

Sliding Window INteraction Grammar (SWING): a generalized interaction language model for peptide and protein interactions, Oral Presentation at: Graduate Research Symposium, November 2024

Sliding Window INteraction Grammar (SWING): a generalized interaction language model for peptide and protein interactions. Oral Presentation at: ISCB GLBio, May 2024

SWING- A generalizable language model for protein and peptide interactions, Oral Presentation at: Center for Systems Immunology Retreat, September 2023

Ionotropic Receptor-dependent warm cells in *Drosophila* larvae, Oral Presentation at: VT School of Neuroscience: Summer Research Retreat, August 2020

Posters:

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. Poster presented at: 25th Annual Immunology Retreat, October 2025.

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. Poster presented at: 23rd Annual Immunology Retreat, October 2023.

Omelchenko A. A., Siwek, J., Chhibbar, P., Rosengart A., Koes D., Joglekar A., Das, J., A generalized language model for predicting perturbations of protein-protein and MHC:Peptide interactions. Poster presented at: Graduate Research Symposium, November 2023.

Omelchenko A. A., Siwek, J., Chhibbar, P., Rosengart A., Koes D., Joglekar A., Das, J., A generalized language model for predicting perturbations of protein-protein and MHC:Peptide interactions. Poster presented at: 21st Annual Immunology Retreat, October 2023.

Omelchenko, A. A., Tyrrell, J. J., Wilbourne, J. T., & Ni, L. Ionotropic Receptor-dependent dorsal organ warm cells mediate warm sensing in Drosophila larvae. Poster presented at: SfN, November 2021; Virtual

Tyrrell, J. J., Wilbourne, J. T., Omelchenko, A. A., Yoon, J., & Ni, L. Ionotropic Receptor-dependent cool cells control the transition of temperature preference in flies. Poster presented at: VT Molecular and Cellular Biology Summer Event, August 2021; Blacksburg, Virginia.

Honors and Awards:

Honors:

- Pittsburgh Magazine 40 under 40 2026 Honoree
- Most Well Rounded Student 2026 – School of Medicine Symposium 2026

Travel Awards:

- St. Jude Travel Award for BioHackathon participation
- BGSA Travel Award for Abstract at ISMB 2025 Conference and Talk
- CRA-WP Grad Cohort For Women Workshop Travel Award 2023
- BrainHack LA Travel Award for 2018 hackathon participation

Presentation Awards:

- Best Abstract – 26th Annual Immunology Retreat 2026
- AAI Trainee Abstract Award- AAI 2026
- Most Technically Impressive Project – St. Jude Hackathon 2025
- Best Abstract – Center for Systems Immunology 2025
- Best Abstract – Graduate Research Symposium 2024
- Best Talk – CPCB Retreat 2024
- Best Abstract – Center for Systems Immunology 2023

LEADERSHIP AND SERVICE

Pennsylvania Woman Work Mentor - Mentoring individuals who are ready for their next career step and need guidance entering the workforce. July 2025-Present

Allegheny County Science Policy Member - Engage in science policy, advocacy, and communication at a local, state, and national level. May 2026-Present

St. Jude HSRI Mentor - Mentored 2 high school interns in the Research Immersion Program. June 2026-July 2026

CPCB Steering Committee - Graduate representative from the CPCB program chosen to meet with the directors across CMU and PITT to steer the progress of the program. September 2024-September 2025

BGSA CPCB Program Representative - Represent the CPCB graduate program in the School of Medicine student government. September 2023- September 2025

CPCB GSA Liaison - Connected the CPCB graduate program to the larger network of the School of Medicine student government. September 2023-July 2025

TechBIO REU mentor - Advised undergraduate students in research ethics as part of the TechBIO REU program. June 2023-August 2023, June 2024-August 2024, June 2025-July 2025

Book Club Mentor - Guided first-year students and facilitated their transition into the graduate program through discussions of assigned readings. September 2023-May 2025

First-Year Ambassador Mentor - Provided assistance and support to incoming students to assist in their acclimation to the graduate school. September 2023-2024

TRAINING

Teaching:

Teaching Assistant. Computational Structural Biology, University of Pittsburgh, *Sept 2023-Dec 2023*

- Developed curriculum, lesson plans, and course materials and led weekly recitations.
- Graded assignments and final projects and held weekly office hours.

Guest Lectures and Invited Talks:

"How to Network at GLBio," invited talk, ISCB GLBio Workshop for Early Career Researchers, May 2024.

"An Overview of Computational Structural Biology," guest lecture, Data to Knowledge course, University of Pittsburgh, June 2023.

Relevant Graduate Coursework:

- Machine Learning for Biomedical Applications (PITT)
- 10-701 Introduction to Machine Learning (CMU)
- Genomics (CMU)
- Cellular and Systems Modeling (PITT)
- Computational Structural Biology (PITT)
- Essential Mathematics and Statistics for Scientists (CMU)
- Computation for Data Science I (Virginia Tech)
- Problem Solving in Genetics, Bioinformatics, and Computational Biology (Virginia Tech)
- Computer aided Drug design (NYU)
- Protein Engineering (NYU)
- Biocatalysts (NYU)
- Biosensors and Biochips (NYU)
- Immunology (NYU)